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ARTICLE



Effects of full windshield head-up display on visual attention allocation

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ABSTRACT

This study explores the impact of full windshield head-up display (FHUD) cues on the visual attention allocation of drivers under different scenarios. Forty-eight participants with driving experience were randomly divided into two groups and asked to drive on a preset route in various virtual driving scenarios created in advance. The full windshield highlights situational cues related to the driving task, such as lane lines, safe vehicle distance warnings, navigation guidance and pedestrian cues. Regarding the perception of situational cues, the number of fixations and mean fixation duration with FHUD were lower than those without FHUD. Furthermore, the dwell time percentage of the driver's forward view with FHUD was larger than that without FHUD, and the dwell time percentage on both sides was smaller than that without FHUD. In conclusion, FHUD may help drivers more effectively perceive cues and improve drivers' visual attention allocation.

Practitioner summary: FHUD may affect drivers' attention while driving. We examined the effect of FHUD on number of fixations, fixation duration and dwell time percentage in the area of interest under different weather scenarios. Experimental results indicated that FHUD could improve drivers' visual attention allocation and help drivers more effectively perceive cues.

Abbreviations: FHUD: full windshield head-up display; HUD: head-up display; HDD: head-down display; AOI: area of interests; AR: augmented reality

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KEYWORDS

Full windshield head-up display; visual attention allocation; number of fixations; mean fixation durations; dwell time percentage

1. Introduction

Driving is a complex perceptual-information-processing task requiring perception, identification, selection, and response processes (Kim and Bishu 2004). Drivers mainly obtain driving information via vision and audition, with vision being used over 90% of the time (Lansdown 1997). Generally, information regarding the environment and vehicle state that drivers receive from visual channels plays an important role in decision-making and driving behaviours (Johnson et al. 2004; Topolšek et al. 2016).

Timely and effective information perception could help avoid errors and accidents (Huebner and Gegenfurtner 2010). Information such as vehicle speed, engine speed, fuel volume, and navigation is shown on the dashboard, termed the head-down display (HDD), which requires drivers to take their eyes off the road. A driver's gaze leaving the road for longer than 2 seconds significantly increases risk (Zwahlen, Adams, and DeBald 1988) and constitutes a factor causing traffic danger (Kazazi et al. 2016). The development of

automobile technology has made a big difference in visual attention allocation during driving. Head-up displays (HUDs) have been developed (Liu 2003; Jakus et al. 2015), that reduce the frequency of drivers checking the HDD (Tretten et al. 2011). HUDs play an auxiliary role in the driver's cognitive decision-making (Eyraud, Zibetti, and Baccino 2015). They are transparent displays that present data without requiring drivers to look away from their usual view. External environment information is integrated into this system; thus, the driver's attention is allocated in a new way (Moore and Gugerty 2010). While driving, the driver needs to observe the instrument display inside and the driving environment outside the vehicle to obtain necessary driving information. However, owing to limitations of human physiological function, when the driver's gaze changes from the vehicle's instrument display in the vehicle to the outside, the eye needs to adapt to changes in sight distance and light intensity. When the contrast between internal and external light is high, the target cannot be

immediately seen. In addition, observing the HDD constitutes a distraction, increasing the difficulty of taking into account the outside environment (Trivedi and Cheng 2007). HUDs help solve these problems and were initially developed for military aviation to reduce the lead time needed to hit a target while manoeuvring. Using an HUD, the pilot could view flight information with the head up so that the pilot could keep looking forward (Charissis and Papanastasiou 2010). Because of HUD's advantages, it has been used in commercial aircrafts and automobiles for enhancing the ability of information perception.

Currently, common HUDs equipped for in-vehicle information displays are small (approximately 4–6 inches), with relatively less information and coverage of the display area and fewer display modes. With the development of augmented reality (AR) and artificial intelligence technology, the concept of a full-windshield HUD (FHUD) has emerged where the whole front windshield is used as the display interface. Through the integrated display of vehicle system information and visual enhancement, FHUDs can assist drivers in obtaining more driving information and reducing the frequency of the driver looking down (Charissis and Papanastasiou 2010). FHUDs can not only integrate and display system information such as speed, engine speed, fuel level, navigation, communication and entertainment in the windscreen, but also visually enhance environment information such as road blocks, road signs, guideposts, road-markings and turning radius based on AR technology (Schall et al. 2013). FHUDs have large information capacity, diverse display modes, and a wide display range (Park and June 2014; Park et al. 2013). HUD research has mainly compared HUD interface design schemes (Gregoriades and Sutcliffe 2018) for determining the relative importance of different information displayed (Beck and Park 2018). FHUD research can also follow this approach.

To better study the effect of FHUDs on drivers' visual attention, we referred to the general model of the allocation of visual attention (SEEV) developed by Wickens et al. (2001), which is based on a bottom-up and top-down process to explain the distribution of visual attention during activity. The SEEV model was initially applied to aircraft piloting (Wickens et al. 2003) and then to vehicle driving (Benedetto et al. 2013). The driver's visual scanning is guided by four basic elements of the SEEV model: salience, effort, expectation and value. Salience, as a bottom-up factor, can be identified by the visual location, size, colour, shape and other factors (Itti and Koch 2001). Effort,

another bottom-up factor, reflects the load on the driver's observation of the instrument interface, which depends on the distance from the eye to an area. Expectation, as a top-down factor, depends on the effectiveness and efficiency of the driver's interactive information feedback. Value, a top-down factor, is the importance of visual elements when performing tasks and the importance of the tasks themselves (Rothkopf et al. 2016).

The simulation scenario is dynamic, and the driver cannot manage all the visual information (Ullman 1984). According to Wickens et al. (2003), visual attention should follow the four basic factors and be guided by top-down factors (expectation and value) for the best result. Conversely, the bottom-up factors (salience and effort) should be minimised. Although a salient element can guide the eye, it will be a visual distraction if it is not relevant. At the same time, a relevant element that requires too much effort is likely to be overlooked. Therefore, optimal allocation of visual attention is highly dependent on the device's design (Eyraud, Zibetti, and Baccino 2015). We can design a device that highlights important information, such as safe vehicle distance and lane line enhancement, but to avoid too much effort from the driver, this information should be placed in the driver's field of vision.

Highlighted cues attract attention (Jonides, 1981) and make drivers respond more quickly (Werneke and Vollrath 2012). We used two AR enhancement techniques to highlight visual information to attract the driver's visual attention. The first is to add information that the road does not have to the road environment, such as turning cues (Kim and Dey 2009). The second emphasises the existing road environment factors, such as pedestrians and deceleration signs (Narzt et al. 2006). These visual cues can help drivers better detect this information, but can also distract the drivers' visual attention.

Owing to the complexity of driving conditions and vehicle performance, the amount of information and the display mode of a full windshield display will directly affect driving efficiency (Tufano 1997; Horrey and Wickens 2004). FHUDs were designed by referring to research results (Beck 2019) on HUD information demand, image visibility and layout. Reasonable HUD information guidance can enhance the driver's subjective situational awareness and information perception level, improve driving performance, alleviate tension and stress, and guarantee vehicle driving safety. On the contrary, it can lead to driver distraction, loss of subjective situational awareness, and cognitive overload and even aggravate the human-

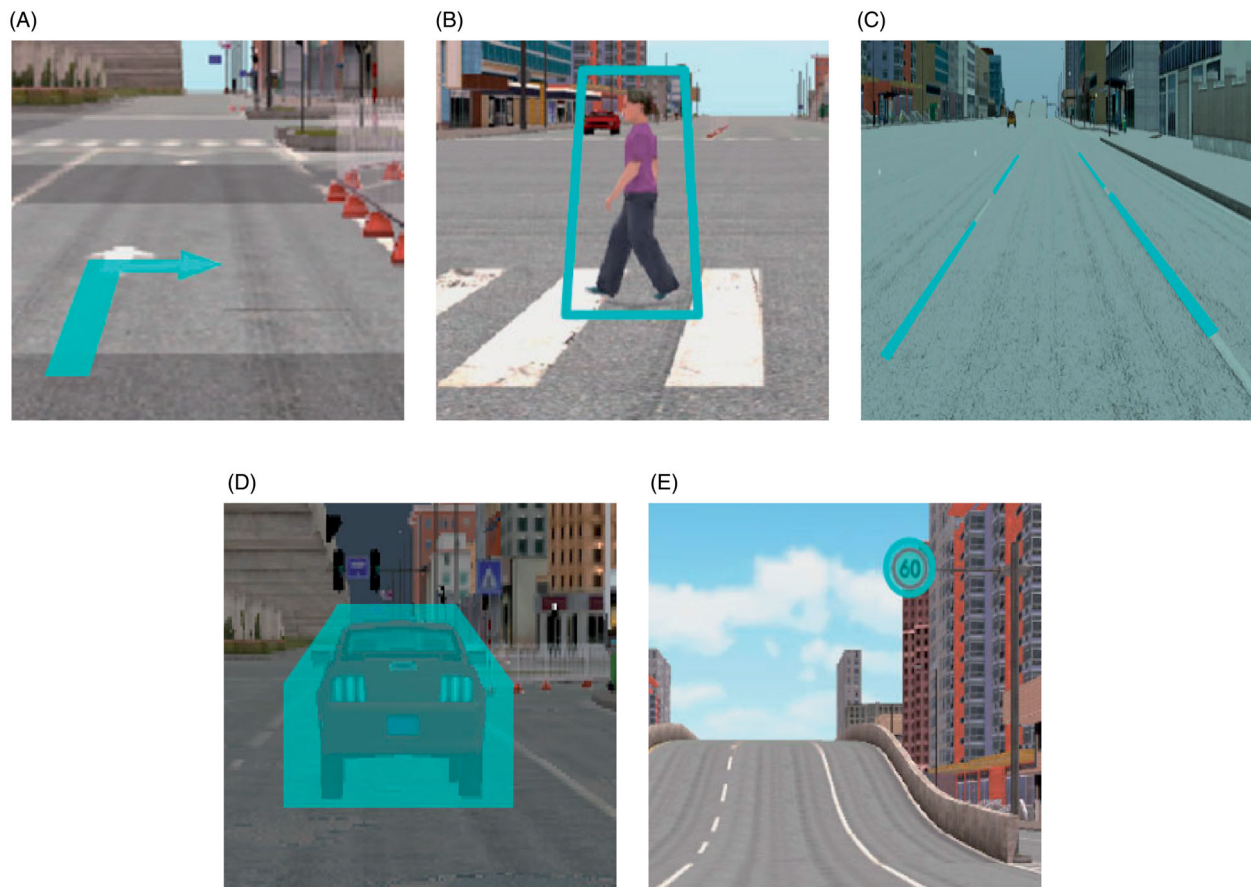


Figure 1. Example to screenshots to demonstrate highlighted cues.

computer interaction security risk of the vehicle (Foyle, Dowell, and Hooey 2001). Therefore, the effects of an FHUD on visual attention allocation is a concern.

Weather is a key factor affecting driving because drivers need to perceive the environment in different weather conditions. Poor weather can negatively impact decision-making ability (Zhang et al. 2005) and lane-keeping ability (Allen, Parseghian, and Stein 1996). Drivers are 3.8 times more likely to show a higher standard deviation of lane position in heavy rain than in clear weather conditions (Ghasemzadeh and Ahmed 2017). To evaluate the effect of FHUDs on visual attention allocation, we analysed differences in visual attention allocation when drivers see relevant information prompts during simulations of sunny, foggy, rainy, snowy, and night scenarios with and without an FHUD.

2. Methods

2.1. Participants

Forty-eight volunteers were recruited to participate in the study [20 women and 28 men; mean $\pm$ SD age = 29 ± 3 (rang 24–36) years] and randomly divided into

two groups. All of participants held a driver's licence and had more than 3 years driving experience. All participants had normal or corrected-to-normal sight and gave their consent to record their eye movements and responses during the study.

2.2. Materials

The visual vehicle driving environment module presents different driving scenarios (sunny, rainy, foggy, and snowy) based on MultiGen and Vega Prime software. In the simulate driving scenario, some cues were highlighted to remind the driver to pay attention to important information (Huang 2013). Highlighted cues were emphasised with semi-opaque shapes superimposed on the image (Figure 1). They were defined and displayed in the simulated scenario in according with traffic rules. Two types of highlighted cues on the screen were defined: road-related and driving-related. The road-related clues were generally related to road conditions (such as traffic lights, landmarks, and pedestrians); driving-related cues were associated with safe driving (such as safe vehicle distance, lane line enhancement, and

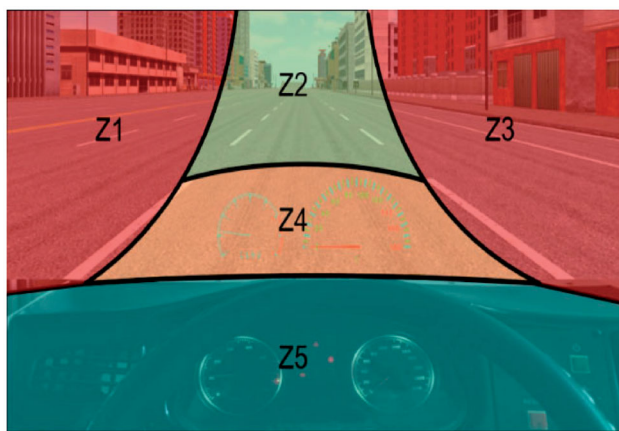


Figure 2. The allocation of each area of interest.

steering cues). In the simulated scenarios, cues were displayed on the FHUD for 3 seconds. Eye-movement data were valid only if there was no obvious change between the driving path and the original path of the driver. The number of highlighted cues in different scenarios differed 13 in the sunny scenario, 15 in the rainy scenario, 16 in the foggy scenario and 14 in the snowy scenario.

The area of interests (AOI) was defined as zones 1–5 (Zhang et al. 2020) according to the driver's visual field and driving environment (Figure 2). Zones 1 and 3 mainly display signs and buildings on both sides of the road as well as speed limit and landmark cues. Zone 2 is mainly displayed in front of the driver's field of vision; for example, safe distance, pedestrian and navigation guidance prompts would be displayed in this area. Highlighted information such as vehicle speed and engine speed is in zone 4, and the dashboard is in zone 5.

2.3. Experimental setup

To investigate the effects of FHUDs on visual attention allocation, a driving simulator used for driver training was modified by the manufacturer. The original simulator includes full-size driving controls found in a normal vehicle. The multi-channel projection system of the simulator presents virtual driving scenarios including sunny, rainy and foggy days. All data interfaces of the modified simulator were available so that real-time data on the vehicle's driving states could be invoked and logged, which provided an important support for the FHUD simulation.

The FHUD interface was developed by using GL Studio software, which generates real-time interactive geometry driven by real-time driving state data received from the simulator data-processing module.

A 60 Hz SMI ETG 2W head-mounted eye-tracking system with a wider tracking range than desktop-fixed ones was used for recording eye movements synchronised with the virtual scenario. Gaze tracking was accurate to 0.5° with a range of 80° horizontally and 60° vertically. The virtual driving scenario was presented on a $4225 \text{ mm} \times 1440 \text{ mm}$ screen. This was expected to provide insight into the direct effect of an FHUD on the driver's attention allocation during driving. Begaze SMI software (version 3.5) was used to analyse eye movement data. Fixation durations were calculated using an algorithm based on the time of eye movement and the minimum duration was set to 100 ms.

2.4. Experimental design

We first used a 4 (type of cue) $\times 4$ (weather) $\times 2$ (FHUD availability) experimental design. The within-subjects independent variables were weather (sunny, foggy, rainy, or snowy) (Figure 3) and FHUD availability (on or off). The between-subjects variable was the type of cue (lane line, vehicle distance, navigation guidance, and pedestrians). Second, we used a 5 (AOI) $\times 4$ (weather) $\times 2$ (FHUD availability) experimental design to establish whether the AOI affected attention allocation. The eye movements of each participant were recorded while driving by an eye-tracking system.

A pre-set route was mapped and provided in advance. The driving routes were geographically the same, and the only difference between routes was weather. The specific order of driving was sunny, rainy, snowy, and foggy. In the same scenario, the driver should first drive without FHUD and then drive with FHUD. All participants were required to drive from the same starting point in each scenario and park the vehicle after arriving at the destination. The connection between the driving route and driver is realised through the navigation arrow cue. Participants encountered variable driving conditions due to the interactive nature of the simulator (e.g. some participants had to stop at a red light, whereas others might have encountered a green light at the same point). The route incorporated a 2×2 two-way urban road with moderate traffic that included traffic lights, right and left turns, and intersections. The driving conditions included other road users that moved normally on the road, obeying traffic laws, and possible overtaking them occasionally. With the FHUD, symbols such as vehicle speed, lane change, lane increase, speed limit, destination, and safe distance maintenance were set to display corresponding to vehicle direction. If the cues were not needed, they would be hidden, and the driver would not notice these



Sunny simulation scenario



Rainy simulation scenario



Foggy simulation scenario



Snowy simulation scenario

Figure 3. Examples of simulation scenarios.



Figure 4. Drivers are experimenting with simulated driving.

cues. Because of the order of routes was not counterbalanced, carryover effects may have existed.

2.5. Procedure

Participants were given a brief introduction to the study and the experimental environment and instructions on how to drive the simulator. All participants drove a 10-minute practice route different from the test route. After the test drive, drivers were given a 5-minute break. The participants then wore and calibrated the eye tracker and began to perform the driving task in sequence (Figure 4). During the entire driving process, they were asked to drive as in their own vehicles and obey all traffic rules.

Eye movements data were recorded by the eye-tracking system. The record started when the

participant began to perform the driving task and stopped when all tasks were completed. Participants were paid after completing all driving tasks.

2.6. Dependent variables

We selected three dependent variables to evaluate the effect of FHUDs on visual attention: AOI dwell time percentage, number of fixations, and mean fixation durations. Dwell time percentage is commonly used to test changes in visual attention allocation. The fixation duration of the AOI is defined as the sum of the fixation durations of each cue during the entire driving process. The number of fixations, defined as a fixed number of times per second, measures the efficiency of information acquisition. Mean fixation duration, defined as the average of the sum of each fixation duration, measures the average time of information acquisition.

3. Results

[A 4 (cue) $\times$ 4 (weather) $\times$ 2 (FHUD availability) mixed analysis of variance (ANOVA) was conducted to identify how FHUD influenced drivers' visual attention. The type of cue was defined as the between-subject factor. The within-subject factors are the weather and the availability of FHUD. The number of fixations and the mean fixation durations of drivers which were the focus of our concerns in the analysis were defined as the dependent variables. The means are reported in Table 1.

Table 1. The means and standard deviations for the number of mean fixations and mean fixation duration of drivers in each zone.

Type of cue	Sun		Rain		Snow		Fog	
	No FHUD	FHUD	No FHUD	FHUD	No FHUD	FHUD	No FHUD	FHUD
Number of fixations								
Lane line	5.5 (0.61)	3.7 (0.63)	6.4 (0.64)	5.5 (0.69)	5.9 (0.59)	4.2 (0.51)	6.3 (0.62)	4.4 (0.54)
Vehicle distance	6.1 (0.81)	4.9 (0.83)	8.3 (0.84)	6.5 (0.89)	8.5 (0.89)	6.9 (0.91)	8.5 (0.92)	6.2 (0.94)
Navigation guidance	2.2 (0.39)	2.1 (0.49)	4.8 (0.41)	3.2 (0.39)	5.5 (0.39)	4.1 (0.41)	6.2 (0.52)	4.3 (0.44)
Pedestrian	2.2 (0.47)	2.3 (0.49)	4.1 (0.39)	3.3 (0.61)	2.5 (0.4)	2.2 (0.42)	3.6 (0.56)	2.9 (0.44)
Mean fixation durations								
Lane line	6.6 (0.81)	4.1 (0.73)	7.5 (0.64)	4.9 (0.69)	9.6 (0.79)	6.6 (0.66)	8.7 (0.62)	5.5 (0.74)
Vehicle distance	7.7 (0.91)	5.7 (0.93)	9.6 (0.94)	6.3 (0.99)	10.3 (0.89)	7.9 (0.91)	14.6 (0.92)	10.2 (0.94)
Navigation guidance	2.3 (0.41)	2.2 (0.43)	4.3 (0.34)	3.6 (0.49)	5.1 (0.39)	4.2 (0.51)	4.1 (0.42)	3.6 (0.44)
Pedestrian	2.4 (0.48)	2.3 (0.43)	4.4 (0.44)	3.4 (0.49)	2.6 (0.41)	2.7 (0.47)	5.1 (0.42)	4.4 (0.44)

The unit of number of fixation is number; the unit of mean fixation durations is second.

In order to study the effect of FHUD on the AOI dwell time percentage. A 5 (AOI) $\times$ 4 (weather) $\times$ 2 (FHUD availability) mixed ANOVA was conducted.

3.1. Number of fixations

This analysis revealed that weather affected the number of fixations [$F(3,764)=34.64$, $MSE = 3.43$, $p < 0.001$, $\eta^2 = 0.12$]. In particular, there were more fixations in the rainy snowy and foggy scenarios than in the sunny scenario (all $p < 0.001$). LSD *post hoc* tests show that there was no significant difference between rainy and snowy ($p=0.098$), foggy and snowy ($p=0.078$), and foggy and rainy ($p=0.912$) in the number of fixations.

The main effect of cue type was also significant [$F(3,764)=374.35$, $MSE = 1.58$, $p < 0.001$, $\eta^2 = 0.60$] with drivers having a more fixation frequency with highlighted lane line cues than with navigation guidance and pedestrian cues (both $p < 0.001$). LSD *post hoc* tests show that there were more fixations on the safe vehicle distance cues than on the highlighted lane line. navigation guidance and pedestrian cues (all $p < 0.001$). There were less fixations on the pedestrian cues than on the navigation guidance cues ($p < 0.001$).

There was a main effect of FHUD availability [$F(1,766)=82.29$, $MSE = 3.51$, $p < 0.001$, $\eta^2 = 0.10$]. With FHUD, drivers had a less fixation frequency than without FHUD.

We noticed an interaction between weather and type of cue [$F(9,736)=47.81$, $MSE = 0.4$, $p < 0.001$, $\eta^2 = 0.37$; Figure 5]. LSD *post hoc* tests show that drivers in the four weather conditions showed more fixation on the lane line cues than on the navigation guidance and the pedestrian cues (both $p < 0.001$) and on the vehicle distance cues than on the navigation guidance and pedestrian cues (both $p < 0.001$).

We also noticed an interaction between weather and FHUD availability [$F(3,736)=18.92$, $MSE = 0.4$, $p < 0.001$,

$\eta^2 = 0.07$]. Four one-way ANOVAs were conducted: one for each type of weather with FHUD availability used as the independent variable. FHUDs affected the number of fixations in the sunny [$F(1,184)=79.25$, $MSE = 1.33$, $p < 0.001$, $\eta^2 = 0.63$], rainy [$F(1,184)=316.03$, $MSE = 0.95$, $p < 0.001$, $\eta^2 = 0.87$], snowy [$F(1,184)=179.85$, $MSE = 1.64$, $p < 0.001$, $\eta^2 = 0.80$], foggy [$F(1,184)=413.51$, $MSE = 1.36$, $p < 0.001$, $\eta^2 = 0.90$] scenarios.

There was an interaction between the type of cue and FHUD availability [$F(3,736)=41.40$, $MSE = 0.4$, $p < 0.001$, $\eta^2 = 0.14$]. Four one-way ANOVAs were conducted, one for each type of cue with FHUD availability as the independent variable. FHUDs influenced the number of fixations on navigation guidance [$F(1,184)=448.01$, $MSE = 0.66$, $p < 0.001$, $\eta^2 = 0.91$], pedestrians [$F(1,184)=25.57$, $MSE = 1.17$, $p < 0.001$, $\eta^2 = 0.36$], lane line [$F(1,184)=342.39$, $MSE = 1.41$, $p < 0.001$, $\eta^2 = 0.88$] and vehicle distance cues [$F(1,184)=169.65$, $MSE = 3.34$, $p < 0.001$, $\eta^2 = 0.79$].

3.2. Mean fixation durations

To assess whether FHUDs affected drivers' cue perception, we analysed fixation duration. Weather affected fixation durations [$F(3,764)=35.86$, $MSE = 7.68$, $p < 0.001$, $\eta^2 = 0.12$]. LSD *post hoc* tests show that fixation durations was more in the rainy, snowy and foggy driving scenarios than in the sunny scenario (all $p < 0.001$), whereas it was less in the rainy driving scenario than in the snowy ($p=0.026$) and foggy ($p < 0.001$) driving scenario. There were more fixations durations in the foggy than in the snowy driving scenario ($p=0.002$).

The main effect of cue type was also significant [$F(3,764)=400.40$, $MSE = 3.41$, $p < 0.001$, $\eta^2 = 0.61$] with drivers having a more fixation duration for highlighted lane line cues than for navigation guidance or pedestrian cues (both $p < 0.001$). LSD *post hoc* tests show that fixation durations were more for safe vehicle

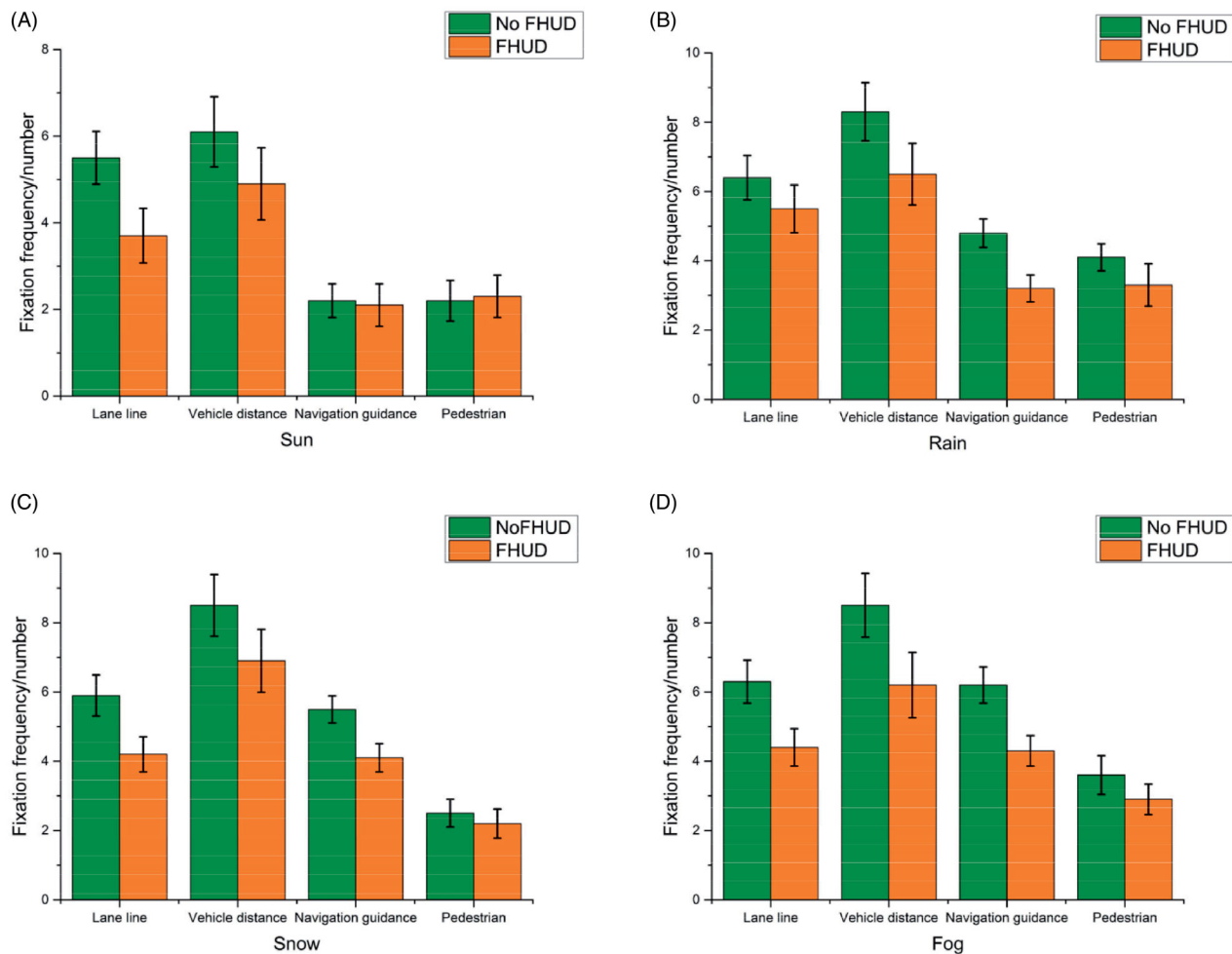


Figure 5. Number of fixations on four different cues in the four simulation scenarios.

distance cues than for the highlighted lane line, navigation guidance or pedestrian cues (all $p < 0.001$). Finally, no significant difference was found between the pedestrian and navigation guidance cues ($p = 0.137$).

There was a main effect of FHUD availability [$F(1,766) = 69.45$, $MSE = 8.01$, $p < 0.001$, $\eta^2 = 0.08$]. Fixation duration was less in drivers with FHUDs than in those without FHUD.

We noticed an interaction between weather and type of cue [$F(9,736) = 143.07$, $MSE = 0.43$, $p < 0.001$, $\eta^2 = 0.64$] (Figure 6). LSD post hoc tests show that drivers in the four weather conditions had more fixation durations for the lane line cues than on the navigation guidance cues and pedestrians (both $p < 0.001$). Fixation durations was more for vehicle distance cues than for navigation guidance cues and pedestrian cues (both $p < 0.001$).

We also noticed an interaction between weather and FHUD availability [$F(3,736) = 22.45$, $MSE = 0.43$,

$p < 0.001$, $\eta^2 = 0.08$]. We performed four one-way ANOVAs. FHUD influenced fixation duration in sunny [$F(1,184) = 140.05$, $MSE = 1.67$, $p < 0.001$, $\eta^2 = 0.75$], rainy [$F(1,184) = 367.30$, $MSE = 2.01$, $p < 0.001$, $\eta^2 = 0.89$], snowy [$F(1,184) = 565.36$, $MSE = 1.77$, $p < 0.001$, $\eta^2 = 0.93$], and foggy [$F(1,184) = 774.78$, $MSE = 1.57$, $p < 0.001$, $\eta^2 = 0.94$] scenarios.

There was an interaction between the type of cues and FHUD availability [$F(3,736) = 220.86$, $MSE = 0.43$, $p < 0.001$, $\eta^2 = 0.47$]. Four one-way ANOVAs were also conducted: one for each type of cue with FHUD availability used as the independent variable. FHUDs influenced fixation duration for navigation guidance [$F(1,184) = 106.77$, $MSE = 0.53$, $p < 0.001$, $\eta^2 = 0.70$], pedestrian [$F(1,184) = 33.20$, $MSE = 1.06$, $p < 0.001$, $\eta^2 = 0.42$], lane line [$F(1,184) = 1525.51$, $MSE = 2.52$, $p < 0.001$, $\eta^2 = 0.93$], and vehicle distance [$F(1,184) = 1766.61$, $MSE = 3.55$, $p < 0.001$, $\eta^2 = 0.91$] cues.

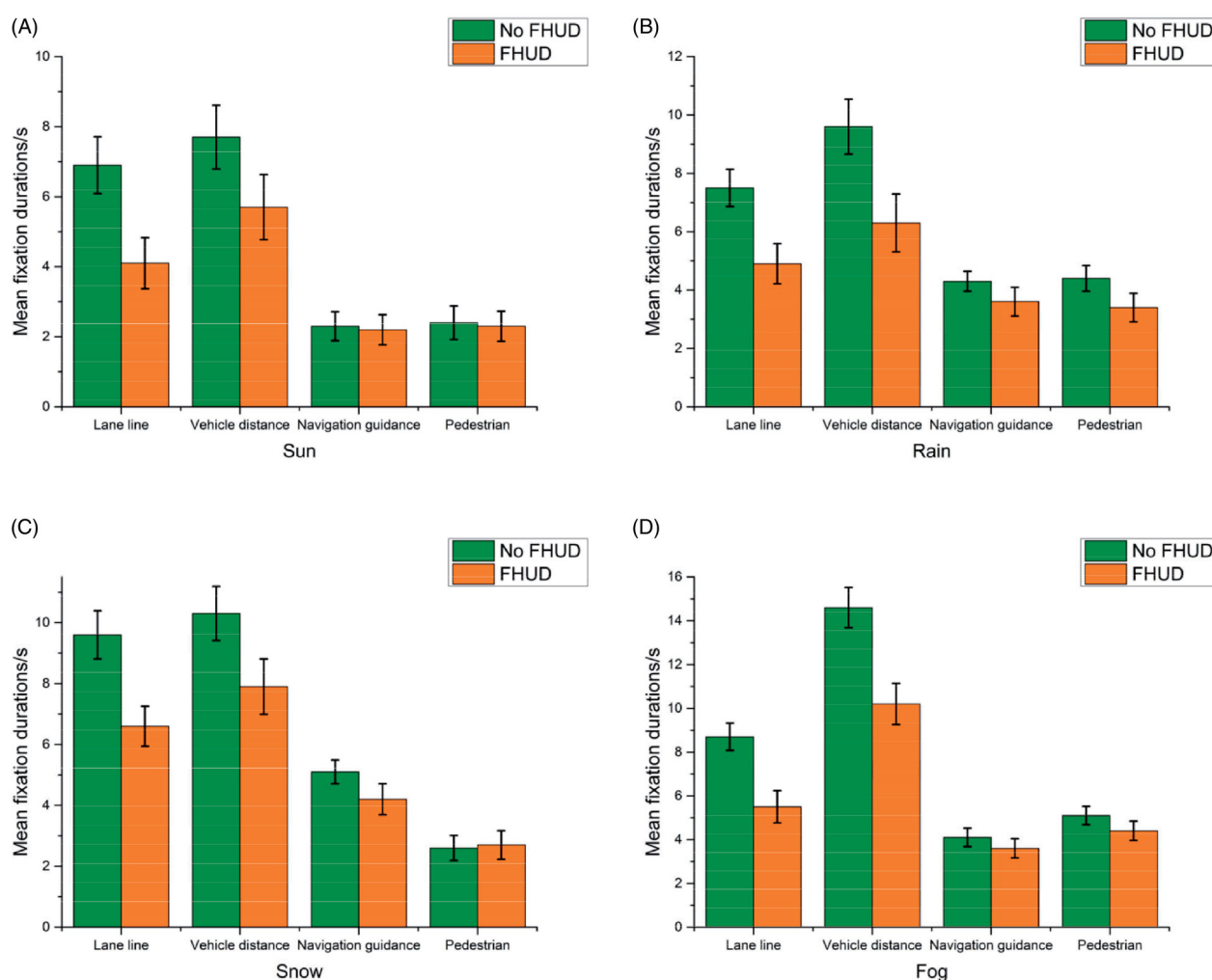


Figure 6. Mean fixation durations on four different cues in the four simulation scenarios.

3.3. Areas of interest analysis

The Figure 7 shows a main effect of the AOI [$F(4,955) = 15.88$, $MSE = 1.54$, $p < 0.001$, $\eta^2 = 0.58$]. LSD post hoc tests show that dwell time percentage was greater in zone 4 than in zone 1 ($p < 0.001$), zone 2 ($p < 0.001$), zone 3 ($p < 0.001$) and zone 5 ($p = 0.02$).

There was an interaction between AOI and weather [$F(12,920) = 6.19$, $MSE = 3.43$, $p < 0.001$, $\eta^2 = 0.63$]. LSD post hoc tests show that drivers with FHUDs had a higher dwell time percentage in zone 4 than drivers had without FHUDs (all $p < 0.001$) for all four weather scenarios.

There was also an interaction between AOI and FHUD availability [$F(4,920) = 21.09$, $MSE = 2.93$, $p < 0.001$, $\eta^2 = 0.53$]. LSD post hoc test shows that dwell time percentage in zone 2 was higher with FHUDs ($p < 0.001$). In the four weather scenarios, drivers' visual attention allocation was similar, but the percentage were higher in drivers with FHUDs (Figure 7) (with vs. without FHUD: 60 vs. 72% in the sunny

scenario; 59% vs. 72% in the foggy scenario; 56 vs. 68% in the snowy scenario; 49 vs. 71% in the rainy scenario. in zone 2 and 4). A lower dwell time percentage was allocated in the bilateral areas (zone 1 and 3).

4. Discussion

This study comprehensively investigated the effects of FHUDs on visual attention allocation. Considering that the weather was an objective factor that cannot be ignored, we simulated four different test scenarios: sunny, rainy, foggy, and snowy. The results showed that FHUDs had affected visual attention allocation. Compared with the control condition, the number of fixations of the drivers in the simulation scenario was lower, as was the mean fixation duration.

It can be seen from Table 1 that there were differences in image enhancement cues such as highlighted lane line, vehicle distance, navigation guidance and pedestrian cues in the two experimental conditions.

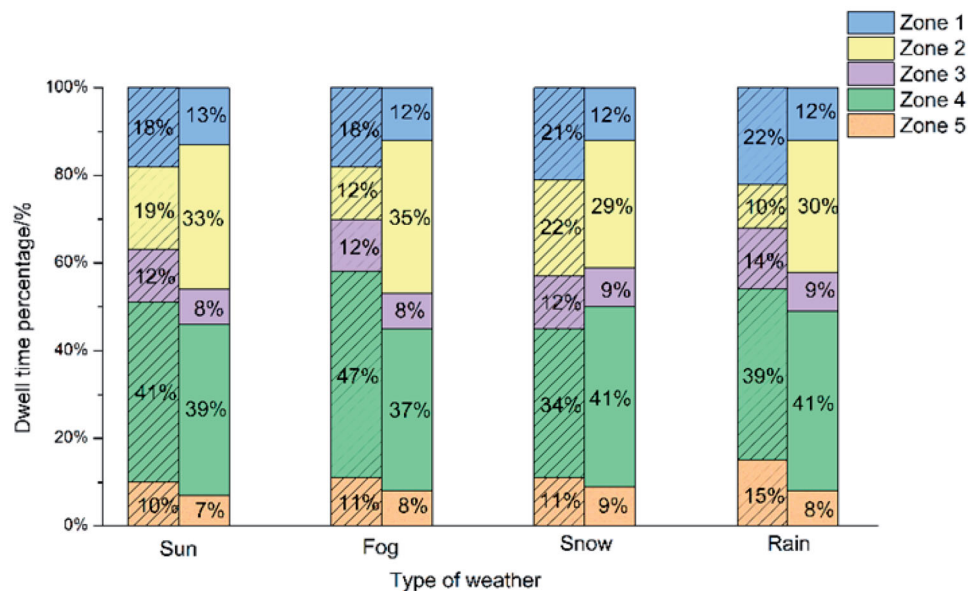


Figure 7. Dwell time percentage of AOI in the four scenarios with FHUD and without FHUD.

Comparing with the control condition, it takes smaller numbers of fixations and shorter mean fixation duration for the drivers to perceive the cues with FHUDs. If we accept that attention is the process of selection of information for further processing, then the results indicates that FHUD could help drivers improve the processing efficiency (Konstantopoulos, Chapman, and Crundall 2010).

A similar conclusion by Liu and Wen (2004), whose research was based on HUD, was also proposed, and they thought that the faster the driver perceives the cue, the shorter the processing time. Our results were similar. FHUD helped drivers perceive the cue faster, but we do not believe that the processing time was shorter. Drivers did not violate the traffic rules during the entire experiment; therefore, we believe that they may have perceived cues successfully and invested the reduced time of perceiving cues into reaction time. This means that the visual attention of drivers was allocated to driving for a longer time, which may decrease the safety risk (Wang et al. 2010). Based on the definition of an AOI as stated in the Methods section 2.2, pedestrian cues appeared in zone 1 and 3, and we found that drivers spent less time on information cues (such as pedestrian cues) on both sides of the road in the control condition. Thus, with FHUDs, the driver can quickly receive processing cues while paying attention to roadside information.

In general, weather conditions affected the drivers' eye movements. As the visibility of the driving environment decreases, the mean fixation duration and processing time of drivers become longer (Charissis,

Papanastasiou, and Vlachos 2008). It can be seen from Table 1 that driving in the low-visibility scenarios (rainy, foggy, snowy weather) makes the number of fixations rise, and so is the mean fixation duration. The results reflect that FHUD plays a positive role when drivers drive in the bad weather.

In the SEEV model (Wickens et al. 2003), according to its eye movement prediction formula, the value of cues is expressed as the product of the correlation of cues and the priority of related tasks. Priority corresponds to the importance of one task compared to another. In the Methods section, we classified the cues into two types: road-related (pedestrians and navigation guidance cues) and driving-related (lane line and vehicle distance cues). It can be seen from the Figures 5 and 6 show that with FHUDs, the number of fixations and the fixation duration for driving-related cues are greater than those for road-related cues in the four scenarios. This indicates that driving-related cues have a higher priority than road-related cues. Therefore, the value of information will be higher when it is related to manoeuvres (Triesch et al. 2003). Drivers can effectively prioritise appropriate cues. Task value has an effect on the driver's visual attention distribution with FHUDs. In bad weather, the number of fixations and the fixation duration of drivers on driving-related and road-related cues are increased. The data show that the number of fixations and the fixation duration in rainy, foggy, and snowy scenarios are higher than those in the sunny scenario. A possible explanation is that drivers increase the amount of time scanning the outside world in bad weather.

Expectancy may influence visual scanning differentially, depending on the type of task and the extent to which ambient vision may be used to support the task (Horrey, Wickens, and Consalus 2006). The expectations of drivers in bad weather are higher than those in sunny weather. To be optimal, salience must be consistent with a high value. Increasing the significance of high-value cues positively affects the driver's attention allocation. The high salience of low-value cues can interfere with the driver's attention. All cues listed in this paper have high values. The results show that FHUDs can reduce the driver's visual attention distribution on high-value cues by salience. We introduced the SEEV model to describe a reasonable and effective visual attention allocation model for the FHUD. Designers can use this model to predict the visual attention allocation of drivers in different weather conditions and thereby predict rationality of cues on the FHUD.

We defined five AOIs for the FHUD, and evaluated drivers' visual attention allocation with FHUD and without FHUD according to dwell time percentage in each AOI (Figure 7). Our results show that the dwell time percentage of zone 2 and 4 were higher than those of other zones under the four experimental conditions. Moreover, when the FHUD is on, the dwell time percentage of zone 2 and 4 are higher than those in the control condition, and the dwell time percentage of zone 1 and 3 are lower than those in the control condition. This shows that the FHUD changes the driver's visual attention allocation, reduces the visual attention allocation of bilateral areas and makes drivers allocate more attention to the front of the field of vision.

5. Conclusion

In this work, the effects of FHUDs on visual attention allocation were investigated. The experiment results showed that FHUDs could help reduce the number of fixations and mean fixation duration needed to observe driving-related and road-related cues in four simulation scenarios. Comparing with the control condition, it takes smaller numbers of fixations and shorter mean fixation duration for the drivers to perceive the cues with FHUDs. For the cues highlighted, drivers allocated more attention on the driving-related cues than the road-related ones. We can also see from the results that FHUDs play a positive role when driving in the bad weather. The AOI analysis showed that the dwell time percentage of zones in front of the driver's field of vision were higher than those of other

zones under the four experimental conditions. Moreover, when the FHUD is on, the dwell time percentage spent in the front are higher than those in the control condition, while the dwell time percentage spent on bilateral areas comparing with those in the control condition is opposite. The AOI analysis indicates that the FHUD changes the driver's visual attention allocation and reduces the visual attention allocation of bilateral areas.

The present study did not evaluate the effect of FHUDs on driving performance but focussed more on visual attention allocation when highlighted cues were provided by FHUDs. We aim to analyse driving performance in future research.

Disclosure statement

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